Example 3.62 Derive cosine formula using the law of sines in a $\triangle ABC$.

Solution:

The law of Sines: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R$

$$\begin{aligned}
\frac{b^2+c^2-a^2}{2bc} &= \frac{(2R\sin B)^2 + (2R\sin C)^2 - (2R\sin A)^2}{2\,(2R\sin B)\,(2R\sin C)} \\
&= \frac{\sin^2 B + \sin(C+A)\sin(C-A)}{2\sin B\sin C} \\
&= \frac{\sin B\,[\sin B + \sin(C-A)]}{2\sin B\sin C} \\
&= \frac{\sin(C+A)+\sin(C-A)}{2\sin C} = \cos A
\end{aligned}$$

Thus, $\cos A = \dfrac{b^2+c^2-a^2}{2bc}$. Similarly we can derive other two cosine formulas.

Example 3.63 Using Heron's formula, show that the equilateral triangle has the maximum area for any fixed perimeter. [Hint: In $xyz \le k$, maximum occurs when $x = y = z$].

Solution:

Let ABC be a triangle with constant perimeter $2s$. Thus s is constant.

We know that $\triangle = \sqrt{s(s-a)(s-b)(s-c)}$

Observe that $\triangle$ is maximum, when $(s-a)(s-b))(s-c)$ is maximum.

Now, $(s-a)(s-b)(s-c) \le \left(\dfrac{(s-a)+(s-b)+(s-c)}{3}\right)^3 = \dfrac{s^3}{27}$ $\quad [G.M \le A.M.]$

Thus, we get $(s-a)(s-b)(s-c) \le \dfrac{s^3}{27}$

Equality occurs when $s-a = s-b = s-c$. That is, when $a = b = c$, maximum of $(s-a)(s-b))(s-c)$ is $\dfrac{s^3}{27}$

Thus, for a fixed perimeter $2s$, the area of a triangle is maximum when $a = b = c$.

Hence, for a fixed perimeter, the equilateral triangle has the maximum area and the maximum area is given by $\triangle = \sqrt{\dfrac{s\,(s^3)}{27}} = \dfrac{s^2}{3\sqrt{3}}$ sq.units.



Exercise - 3.9

1. In a $\triangle ABC$, if $\dfrac{\sin A}{\sin C} = \dfrac{\sin(A-B)}{\sin(B-C)}$, prove that a^2, b^2, c^2 are in Arithmetic Progression.
2. The angles of a triangle ABC, are in Arithmetic Progression and if $b : c = \sqrt{3} : \sqrt{2}$, find $\angle A$.